DT081A — Laboratory 3 Report

Student: Jakob Obid

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1 Introduction

This lab studies how pixel values, colour channels, filters and spatial patterns affect digital images.

2 Experimental Setup

I used Python, NumPy, OpenCV and Matplotlib. The main test image was `ManWoman.png`, with 631×475 pixels. For the first experiment, I converted it to grayscale and applied gamma correction.

3 Results and Analysis

3.1 Human Perception of Color, Spatial Detail, and Gamma Correction

Q1.1. Implement gamma correction and calculate the outputs for intensities 64, 128 and 192 using gamma = 2.0.

Answer: I used gamma correction with gamma = 2.0. The outputs were 16, 64 and 144 for the three input values. The values were converted to integers for the final result.

Input	Output
64	16
128	64
192	144

Q1.2. Compare the image histograms before and after applying gamma = 2.2. Explain the relationship to nonlinear light perception.

Answer: I applied gamma = 2.2 to the grayscale image. The mean intensity fell from 95.28 to 46.25, and the histogram moved toward darker values (Figure 1). This shows how a nonlinear intensity change affects brightness, which relates to how the eye responds nonlinearly to light.

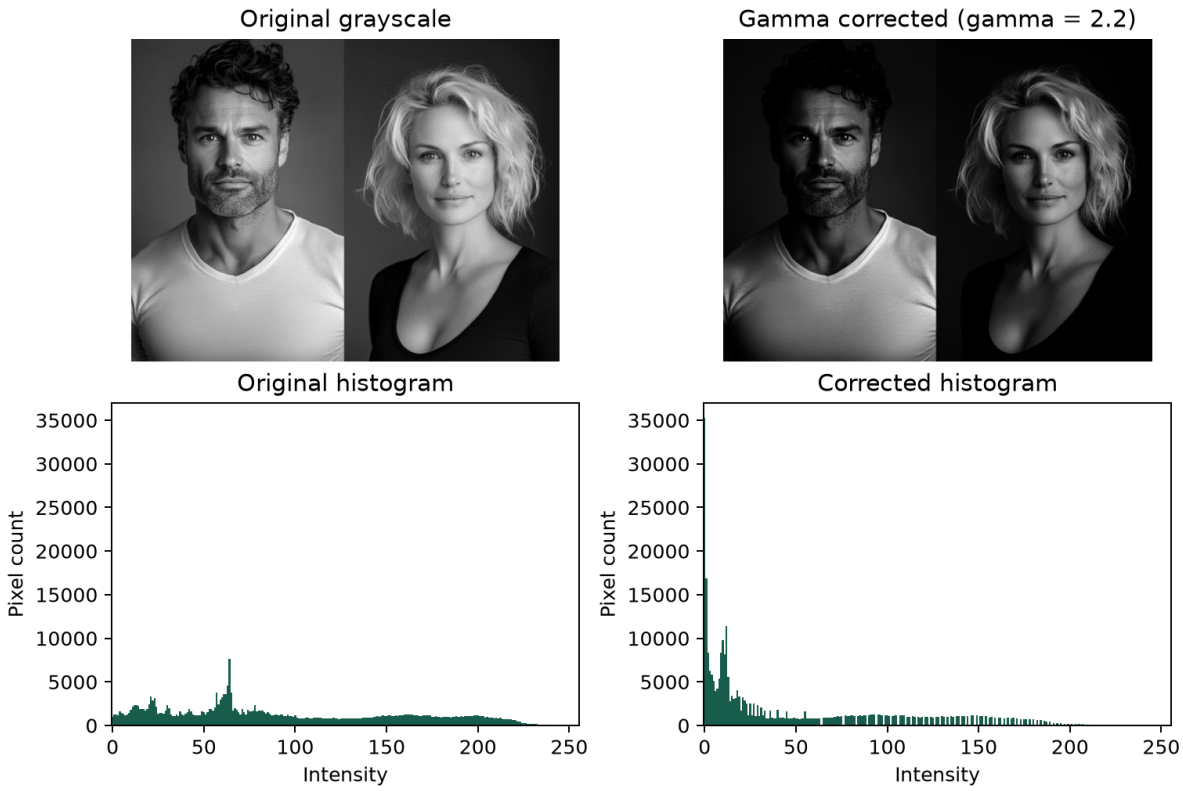


Figure 1: Original grayscale image and gamma-corrected image (gamma = 2.2), with their intensity histograms.

4 Discussion

Gamma correction made the image darker because gamma was greater than 1.

5 Conclusion

The first experiment showed how changing gamma affects image brightness and its histogram.